

Learned Spacecraft Pose Estimation: Failure Detection and Uncertainty

Experiments on monocular relative pose estimation for rendezvous and proximity operations

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1. Problem statement

Autonomous rendezvous and docking relies on active sensing and cooperative targets. Learned monocular pose estimation has been proposed [2], [6] as a way to extend that capability to unprepared targets and degraded illumination, but it is not yet flown. Three barriers are identified consistently in the literature review of multiple studies: the synthetic-to-real domain gap, deployment inside a flight processor and its budget, and predictable behaviour when an estimate is unreliable.

The following experiments explore and approach the 3 problems mentioned above. The question throughout is, not which model is most accurate, but **whether the perception stage can determine that it is wrong, at runtime and without ground truth**. A pose returned with high confidence that is in fact wrong, propagates unchecked into the navigation filter. A pose flagged as uncertain can be rejected, retried or cross-checked against another sensor. Accuracy and self-assessment are therefore separate requirements, and this assessment is the one that decides whether a learned component can sit inside an engineering navigation architecture.

2. Experiments performed

S/N	Title	Data	Principal source
1	Keypoint detector benchmark on clean imagery (ORB, SIFT, DISK)	Synthetic wireframe	Kisantal et al. 2020 [3]; Tyszkiewicz et al. 2020 [7]
2	Robustness under controlled degradation	Synthetic wireframe	Pauly et al. 2023 [6]
3	Failure detection and confidence calibration	Synthetic wireframe	Guo et al. 2017 [8]
4	Convolutional and transformer backbones; measurement of the domain gap	SPEED+ v2	Park et al. 2022 [4]; Oquab et al. 2024 [9]
5	Aleatoric and epistemic uncertainty across the domain gap	SPEED+ v2	Kendall & Gal 2017 [10]; Gal & Ghahramani 2016 [11]
6	Confidence-driven measurement covariance in an EKF; Monte Carlo campaign	Simulated HCW trajectory	Martinez et al. 2023 [1]
7	Deployment cost: parameters, operations, latency, quantisation	Both data models	Cosmas & Asami 2020 [12]; Posso et al. 2024 [13]

Experiments 1 to 3 share one notebook and one testbed. Experiments 4 and 5 share the SPEED+ data and the same network architecture, trained separately. Experiment 6 is the filter study, Experiment 7 the deployment study. All code and executed notebooks are in the [github repository](#).

3. Experiments 1 - 3: detector benchmark, robustness and calibration

Idea: The hybrid architecture used here, is inspired by the winning solutions of the first Satellite Pose Estimation Competition, SPEC2019 [3]. A network locates known structural keypoints, and a Perspective-n-Point solver with RANSAC recovers the pose geometrically. That separation keeps the geometry analysable. The experiment answers how the choice of keypoint detector propagates through it, and whether the solver's own by-product, the RANSAC inlier ratio, can serve as a runtime confidence signal.

Method. A programmatic spacecraft-like wireframe with 16 keypoints and exact ground truth, rendered at 40 random poses. Three detectors: ORB and SIFT as classical baselines, and DISK [7], a pretrained convolutional detector and descriptor, as the learned component. Four degradations (motion blur, brightness, additive noise, occlusion) at six severity levels, giving 2,880 runs. Failure is defined as translation error above 1 m or rotation above 10°. Calibration is measured with Expected Calibration Error [8].

Detector	Detections	Matched (of 16)	Transl. (m)	Rot. (°)	ECE	ms/frame
ORB	585	16	0.092	0.77	0.091	6.2
SIFT	44	7	0.176	4.57	0.188	73.7
DISK (CNN)	1024	9	0.232	0.50	0.232	72.7

Table 1. Detections, matches and pose error on one example clean frame; ECE over all 2,880 runs; time per frame (ORB and SIFT on CPU, DISK on a Colab T4 GPU). DISK has 1.09 M parameters.

Result. Detection count does not predict usefulness: DISK produced the most detections but the lowest fraction on model keypoints (9 of 1,024 on the example frame). Accuracy and calibration are separate: across all 2,880 runs DISK had the worst-calibrated confidence, ECE 0.232 against 0.091 for ORB. When its inlier ratio reported high confidence, the pose was wrong considerably more often than that value implied. The degradations separated the detectors in physically explainable ways: motion blur degraded ORB first, since the FAST corner response depends on high-frequency edge energy. Occlusion degraded all three detectors together once the occluder covered 40% of the image or more.

Limitation. Association in these experiments used the ground-truth projections of the model keypoints. This is a reasonable model of a cooperative target carrying identified markers, as in the InSPoC-1 architecture [1], but not of an unprepared one, and it makes absolute figures optimistic. The comparison between detectors is unaffected.

4. Experiment 4: backbone comparison and measurement of the domain gap

Idea: Experiments 1 to 3 used only rendered imagery, so they did not measure the gap between rendered images and images of physical hardware. SPEED+ [4] dataset was used for this experiment: 59,960 labelled synthetic images of the Tango spacecraft, plus two hardware-in-the-loop domains photographed in Stanford SLAB's TRON facility; lightbox, simulating Earth albedo, and sunlamp, simulating direct sunlight. The SPEED+ v2 dataset includes labels for both, so the Domain gap can be measured rather than inferred. The second question is whether a transformer backbone helps, since attention is one of the directions under consideration.



Figure 1: the visual difference between the three domains: evenly lit synthetic images, Earth in frame on lightbox, saturation and stray light on sunlamp

Method. Two backbones are tested under an identical head, loss and schedule. A fully fine-tuned ResNet-18 (11.57 M parameters, all trained) and a frozen DINOv2 ViT-S/14 [9] (22.58 M parameters, 0.53 M trained in the head). 8,000 synthetic training images, 12 epochs, evaluated on 1,500 held-out synthetic and 600 images per hardware-in-the-loop domain. No domain adaptation was applied, deliberately, so that the untreated gap is what is observed.

Model	Domain	Transl. (m)	% of range	Rot. (°)	Factor
ResNet-18	synthetic	0.271	5.07	27.5	1.00
ResNet-18	lightbox	3.920	64.19	133.4	14.47
ResNet-18	sunlamp	4.174	66.64	132.8	15.41
DINOv2 ViT-S/14	synthetic	0.366	6.33	52.9	1.00
DINOv2 ViT-S/14	lightbox	2.100	36.26	107.9	5.74
DINOv2 ViT-S/14	sunlamp	1.591	26.46	106.9	4.35

Table 2. Median pose error by domain. 'Factor' is degradation relative to the synthetic domain.

Result. The ranking reverses across the domain gap. The ResNet backbone is more accurate on synthetic imagery (0.271 m vs the Transformer's 0.366 m). However, the ResNet performs significantly worse on the hardware-in-the-loop imagery. Its degradation factors are 14.5 and 15.4, while the transformer is much more robust, degrading by factors of only 5.7 and 4.4. The mechanism behind this failure is visible in the predicted range distributions: on both hardware-in-the-loop domains, the ResNet-18 collapses onto a narrow peak near 2.2 m and returns nearly the same answer regardless of image content, while the transformer successfully retains a realistic spread.

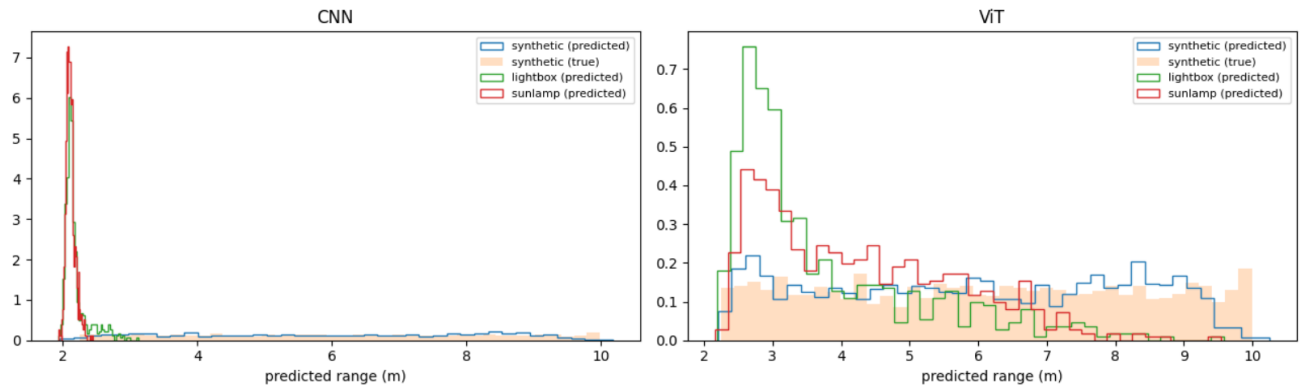


FIGURE 2: the collapse of the ResNet-18 to a near-constant prediction on hardware-in-the-loop imagery, against the retained spread of the DINOv2 transformer

Error is strongly anisotropic. On the synthetic validation set, both models struggle significantly more with depth than with lateral positioning. For the ResNet-18, the median error along the camera's boresight (depth) is 0.258 m compared to just 0.061 m across it (lateral), a factor of 4.2. The transformer shows a similar but less extreme trend (0.297 m vs. 0.138 m, a factor of 2.2). This discrepancy occurs because a single camera easily determines lateral position based on where the target appears in the frame, but must infer range purely from how large it appears. This is the same observability argument behind the laser rangefinder that provides range in the cooperative architecture of [1].

The necessity of domain adaptation. Neither model is usable on untreated hardware-in-the-loop imagery. Both exhibit severe degradation, with position errors reaching 26% to 67% of the total range and rotational errors exceeding 100°. This aligns with the SPEC2021 analysis [5], which concluded that competitive solutions heavily depended on domain adaptation using target-domain imagery.

Training cost vs. trainable parameters. having fewer *trainable* parameters did not equate to faster training. Even though the transformer's backbone was frozen, it took 42 seconds per epoch compared to the fully fine-tuned ResNet's 24 seconds. Freezing weights removes the need for a backward pass through the backbone, but the computationally expensive forward pass remains unchanged.

5. Experiment 5: aleatoric and epistemic uncertainty

Idea. The confidence signal in Experiment 3 is generated by a solver and has no units. But, a navigation filter requires a calibrated covariance matrix. This experiment trains a network that predicts its own error and separates it into two components: aleatoric uncertainty, which is noise in the observation and is predicted by a variance output trained with a Gaussian negative log-likelihood [10]; and epistemic uncertainty, the model's own ignorance, estimated by Monte Carlo dropout [11]. Epistemic uncertainty requires no labels, so it is a candidate mechanism for detecting a domain shift in flight.

Result on synthetic domain. Predicted aleatoric standard deviation is [0.098, 0.123, 0.882] m per axis and epistemic is [0.048, 0.070, 0.656] m. The boresight component is about 7 to 9 times the lateral ones for aleatoric and 9 to 14 times for epistemic. The loss is a per-axis Gaussian likelihood, so the model sees per-axis residuals but is never told which axis is harder. This is the same direction as the 4.2x anisotropy measured in Experiment 4, though larger, and it comes from a different trained network. The model's calibration is conservative: the median ratio of actual error to predicted standard deviation is 0.40, against about 1.0 expected for calibrated error bars with this anisotropy, so the predicted error bars are roughly two and a half times wider than the observed errors. The correlation between actual error and predicted standard deviation is +0.326. The two components are not independent: per image, epistemic and aleatoric are close to linearly related, so the split separates the arithmetic more than two distinct sources.

Domain	Aleatoric (m)	Epistemic (m)	Epistemic ratio	ROC area
synthetic	0.367	0.258	1.00	—
lightbox	0.132	0.101	0.39	0.076
sunlamp	0.107	0.148	0.57	0.203

Table 3. Mean per-axis predicted standard deviation by domain, and ROC area for separating synthetic from hardware-in-the-loop images using the epistemic term alone.

Result across the Domain gap: Epistemic uncertainty does not detect the domain gap. It moves the wrong way. Mean epistemic standard deviation is 0.258 m on synthetic against 0.101 m on lightbox and 0.148 m on sunlamp, ratios of 0.39 and 0.57, and aleatoric falls similarly, to ratios of 0.36 and 0.29. The total covariance this network would hand to a filter therefore shrinks on the images it handles worst: its own median translation error on lightbox and sunlamp is 13.4 and 13.9 times its synthetic error. Scored as a detector, the areas under the ROC curve are 0.076 and 0.203, both below the 0.5 of an uninformative signal.

Mechanism: prediction collapse. On the hardware-in-the-loop images the spread of this network's predicted range shrinks to 0.16 m and 0.11 m, against 2.24 m on synthetic images: it returns nearly the same answer for every image, the same collapse Experiment 4 observed. Monte Carlo dropout measures disagreement between stochastic passes; when the output has collapsed, the passes agree and the epistemic estimate is low. Prediction collapse presents as confidence. While ROC areas well below 0.5 mean the signal becomes informative if inverted, an inverted detector cannot be trusted on a domain against which it has not been calibrated, which is the only scenario where it would be useful.

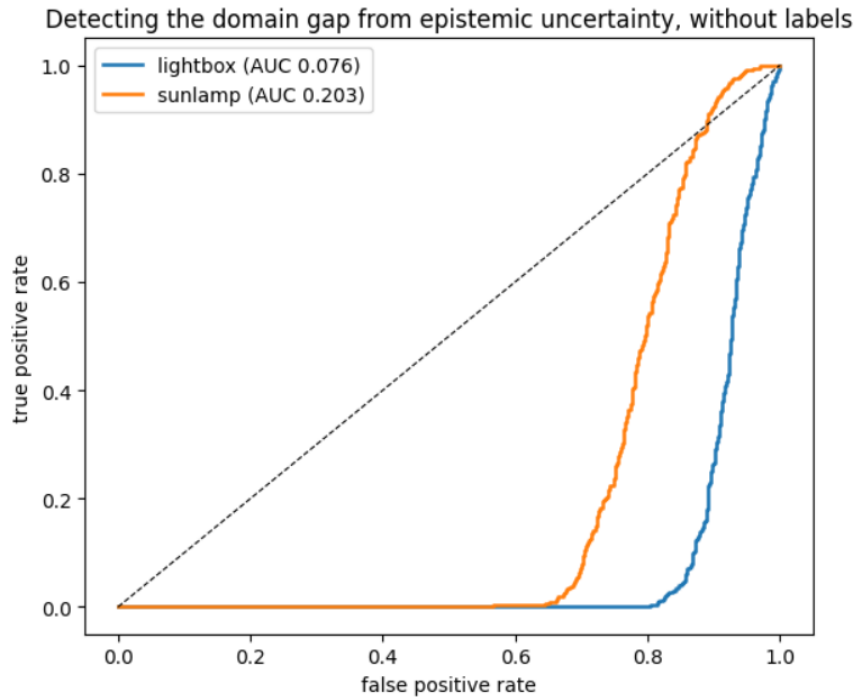


FIGURE 3: ROC curves falling below the diagonal, areas of 0.076 (lightbox) and 0.203 (sunlamp) rather than the 0.5 of an uninformative signal

6. Experiment 6: does a confidence signal help the navigation filter?

Idea. Experiments 1 to 5 evaluate performance on single, static images. But, a navigation system runs a filter over a sequential trajectory. What a navigation filter requires from the perception system is, a measurement together with a measurement covariance R , which determines how aggressively the state estimate moves towards each new measurement. A confidence signal is only useful if it can dynamically and accurately set R . To test this, the dynamics, filter structure, campaign style, and reporting format follow the cooperative rendezvous architecture detailed in [1].

Method. The simulation models a V-bar approach from 20 m using Hill-Clohessy-Wiltshire (HCW) relative dynamics. Measurements are produced by running Experiment 1 pose chain with the ORB detector on rendered frames at 1 Hz. They feed a Kalman filter on relative position and velocity; the measurement is linear in the state, so the filter is written in EKF form but is linear here. The measurement bias and the confidence-to- R table are estimated on 10 separate calibration runs (540 frames), not on the Monte Carlo runs. The campaign consists of 60 Monte Carlo runs of 100 steps with randomised degradation profiles. Consistency is scored with the **Normalized Estimation Error Squared (NEES)** on the three position axes; for a consistent filter it follows a chi-square distribution with 3 degrees of freedom, whose median is 2.37.

The symmetry problem (a geometric failure, not a tuning artefact). The 16 keypoints of the target spacecraft are exactly symmetric under a 180° rotation about its z axis. Mathematically, rotating the spacecraft 180 degrees maps the keypoints perfectly back onto themselves with zero residual error. A method that uses only these untextured keypoints cannot distinguish the two interpretations from one frame; texture or individually coded markers would. To expose the filter to this failure, the symmetric association is injected at a controlled rate (12% of frames) and the real PnP solver is run on it, so the confidence reported for the flipped hypothesis is what the solver genuinely returns.

	Correct association	Flipped association	Difference
Median position error	0.069 m	0.069 m	none
Median attitude error	0.76°	179.9°	179°
Mean inlier ratio	0.908	0.906	0.002

Table 4. The symmetry failure corrupts attitude while leaving position and confidence unchanged. 90% of flipped frames still report confidence above 0.8.

Configuration	RMSE (m)	Final error (m)	Median NEES	Mean NEES	Used
Fixed R (no bias correction)	0.434	0.039	8.88	10.03	79%
Fixed R (bias corrected)	0.439	0.052	3.07	7.54	79%
Adaptive R	0.437	0.048	3.11	7.06	79%
Gated (floor 0.35)	0.437	0.048	3.11	7.06	79%

Table 5. Monte Carlo campaign, 60 runs. For a consistent filter the median NEES is 2.37.

Result. On flipped frames the median position error is 0.069 m, identical to correct frames, while the attitude error is 179.9 degrees against 0.76 degrees. The mean inlier ratio is 0.906 on flipped frames and 0.908 on correct ones, and 90 per cent of flipped frames still report a confidence above 0.8. A geometric confidence signal reports how well a hypothesis was fitted; it cannot report that the wrong hypothesis was fitted well. Because the filter estimates position only, this failure does not reach the filter; it matters for any function that uses the attitude.

The inlier ratio is barely correlated with position error (correlation -0.09), so setting R per frame from it gives the same result as a single fixed value. No measurement fell below the 0.35 confidence floor, so the gated and adaptive configurations are identical and the gated row adds nothing. The measurements carry a systematic offset of [-0.0078, 0.0378, -0.0044] m on the R-bar, V-bar and H-bar axes; on the R-bar axis it is larger than the measurement's own standard deviation. Without correction the median NEES is 8.88. Subtracting the bias estimated on the calibration runs brings it to 3.07, against 2.37 for a consistent filter. Bias correction changes what the filter believes about its error, not the error itself, so the RMSE is essentially unchanged (0.434 m against 0.439 m). The RMSE is larger than the error of a single measurement because it is dominated by the far part of the approach: beyond about 9 m the spread of the filter error across runs reaches several metres, while closer in it is much smaller. The mean NEES stays well above the median (7.54 against 3.07), so some runs remain inconsistent after the correction; examining those runs individually is the next check.

The measurement covariance table shows the along-track component, which is range in this geometry, roughly eight times larger than the two lateral axes. The same ordering appears in the SPEED+ synthetic error of Experiment 4 and in the predicted uncertainty of Experiment 5.

7. Experiment 7: deployment cost

Method. Accuracy alone does not decide model deployment. Rendezvous navigation runs entirely onboard the spacecraft, since ground contact is intermittent and too slow for closed-loop control during the critical minutes it takes to traverse the docking corridor. Furthermore, representative space-flight processors are radiation-hardened and have no GPU. A modern CPU is two to three orders of magnitude faster than a GR740, so the single-thread CPU figure below should be viewed as an optimistic proxy. Thus, achieving flight-rate inference usually requires dedicated hardware acceleration [12][13] rather than simply changing the software architecture.

Model	Params	Size	GMACs	GPU	CPU 1-thread	MC head-only	MC full
ResNet-18	11.57 M	46.3 MB	1.74	3.3 ms	53.6 ms	76 ms	1366 ms
DINOv2 ViT-S/14	22.58 M	90.4 MB	6.12	7.9 ms	136.6 ms	192 ms	3708 ms

Table 6. Per-frame cost at batch size 1. GPU: Colab T4, CUDA synchronised. CPU columns: one thread, measured together in one cell. MC head-only: backbone once, head 25 times. MC full: whole network 25 times.

Result. For a single forward pass, both models fit within the assumed budget: with a 1 Hz camera and half of each interval (500 ms) allocated to perception, both use well under half of that allocation (Table 6).

The cost of the uncertainty estimate depends on where the dropout sits. With dropout only in the head, the backbone runs once and 25 head passes cost about 1.4 times a single forward pass (76 ms against 54 ms for the CNN, 192 ms against 137 ms for the ViT) and still fit the budget. Running the whole network 25 times, as dropout inside the backbone would require, takes 1366 ms and 3708 ms, 2.7 and 7.4 times the 500 ms budget. The cheap estimate is the one Experiment 5 found to be inverted; the stronger estimators are the ones that do not fit the budget.

Quantisation. Dynamic INT8 quantisation converts linear layers only. It shrinks the ViT by 3.59 times, from 90.4 MB to 25.2 MB, but the CNN by only 1.03 times, because almost all of the ResNet-18 arithmetic is in convolutions. Accuracy

after quantisation was not measured on the trained networks; the ViT would normally need quantisation-aware training, and the CNN static quantisation of its convolutions.

8. Synthesis and Conclusion

Four confidence mechanisms were evaluated across two targets and two datasets. All four fail in the same direction: **they report high confidence where the system is most wrong.**

Exp.	Mechanism	Observed failure
3	RANSAC inlier ratio	Learned detector worst calibrated of three, ECE 0.232 against 0.091
4	Predicted range distribution	Collapses to a near-constant value on hardware-in-the-loop imagery; error rises 14.5x and 15.4x
5	Epistemic uncertainty	Falls to 0.39x–0.57x where this network's own error rises 13.4x–13.9x; ROC area 0.076 and 0.203, inverted
6	Inlier ratio inside an EKF	Identical on frames wrong by 180° in attitude; filter inconsistency traced to an unmodelled measurement bias

Table 7. Observed failures of each method

None of these failures is a tuning artefact. In Experiment 6 the attitude failure follows from an exact geometric symmetry of the keypoint set. In Experiment 5 the inverted epistemic signal follows from prediction collapse, measured on the same network.

The Observability Problem:

Beyond confidence estimation, a consistent secondary finding emerges regarding depth observability.

The along-boresight(range) axis is identified as the system's weakest link three separate times:

- (A) 4.2x larger error than the lateral axes on the SPEED+ synthetic validation set (Experiment 4).
- (B) 7 to 14 times larger in the model's own predicted uncertainty (Experiment 5).
- (C) Roughly eight times larger in the navigation filter's measurement covariance (Experiment 6).

All three were measured on synthetic or rendered imagery. The result is consistent with the observability argument for taking range from a laser rangefinder rather than from a monocular camera, as in the cooperative rendezvous architecture of [1].

9. Limitations of this study:

- **Training Budget:** Training used 8,000 of 47,966 available SPEED+ synthetic images for 12 epochs (Experiment 4) and 15 epochs (Experiment 5). Absolute errors are therefore high, and well above published SPEC2021 competition leaderboards; The objective here was to compare two backbones under an identical, constrained computational budget.
- **Direct Regression:** Attitude is regressed directly as a quaternion, with no keypoint or geometric stage. This is the weaker of the two standard approaches and shows in the rotation errors.
- **No domain adaptation:** Domain adaptation was deliberately omitted to ensure the untreated domain gap was what was actually being measured and reported.
- **Asymmetric Backbone Training:** The transformer backbone is frozen, so the comparison is a fine-tuned convolutional network against a frozen self-supervised representation with a trained head.

- **Uncertainty Estimation Limits:** Monte Carlo dropout is a cheap approximation to a posterior, and dropout is present only in the network head. A deep ensemble is the stronger estimator, so the negative results in Experiment 5 do not answer the question for epistemic uncertainty in general.
- **Simulated Trajectories:** Experiment 6 uses a simulated trajectory and rendered imagery, since no public dataset provides labelled rendezvous sequences at the scale a Monte Carlo campaign requires.
- **Hardware Proxies:** Deployment figures come from general-purpose hardware. Power and radiation tolerance are not measured.
- **Measurement bias:** the Experiment 6 bias is estimated once, from calibration runs of the same simulator. A flight system would need to estimate it from ground calibration or online.

10. Proposed direction

These experiments explore possible solutions to develop a failure detection mechanism for learned pose estimation that remains trustworthy. The following are the possible directions that can be explored to solve this problem.

- **Apply domain adaptation** and establish whether uncertainty behaviour improves once the gap is treated rather than merely observed, which is the condition under which a learned front end could be considered for a filter.
- **Replace Monte Carlo dropout with a deep ensemble** and test whether the inversion observed in Experiment 5 persists, or whether it is specific to dropout applied to the head alone.
- **Adopt keypoint regression followed by PnP** instead of direct pose regression, so that geometric consistency provides a check independent of the network's own confidence. This is the architecture of the top-ranked entries in SPEC2019 [3], and the one closest to the classical chain already in operational use [1].

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